GeoPops demonstration: An open-source, adaptable framework for agent-based modeling on synthetic populations

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GeoPops

- **Open-source package for generating geographically and demographically realistic synthetic populations for any US Census location using publicly available data**
- Same methodology as GREASYPOP-CO (One Health Trust) but wrapped in convenient Python package that can be pip installed
- Automatic data downloading
- Compatibility with recent Census data (developing)
- Compatibility with ABM software Starsim (IDM)

github.com/ACCIDDA/GeoPops

```
pip install geopops
```

MIDAS Notebook Tutorial

Demonstration of framework

1. Make a population of Howard County, MD
2. Run SIR model of generic upper respiratory infection with Starsim
3. Incorporate prevalence-dependent "school closures"
4. Track outcomes by age and Census Block Group (CBG)

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

Output: Agents and networks

Agent attributes
- Age
- Gender
- Race/ethnicity
- Worker status
- Student status
- School grade
- Commuter status
- Income category
- Workplace category
- Home CBG
- School CBG if applicable
- Work CBG if applicable
- GQ CBG if applicable

School Network
- List of students for each
- Size of grade levels
- Number of teachers
- CBG locations

Household Network
- List of agents for each
- CBG locations

Workplace Network
- Number of workers
- CBG if inside population area

Group quarters (GQ) Network
- Number of residents
- Number of staff
- CBG locations

- Common network structure when modeling respiratory infections
- Few open-source methods for making this and fewer with spatial data

Methods: Generating agents and households

Public Use Microdata Samples (PUMS)

- Subset of census survey responses
- Compositional details, e.g., the number of 5-member households in a CBG with school-aged children

Combinatorial optimization (CO)

American Community Survey (ACS)

- Demographic data at fine geographic scale
- Limited to marginal counts, e.g., the number of households in a CBG with 5 members and the number of households in a CBG with school-aged children

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698

Methods: Assigning agents to schools and workplaces

	Data source	Data	Methods
Schools	National Center for Education Statistics (NCES)	Student and staff counts for each grade level in public schools	• Students assigned to closest school that offered required grade • Teachers assigned to school by selecting teachers from nearby CBGs
Workplaces	Longitudinal Employer-Household Dynamics (LEHD)	Origin-destination and workplace characteristics	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers

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Methods: Generating Contacts

Layer	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: α=0.9 • Mean degree: K=12 • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: α=0.9 • Mean degree: K=8 • Blocks are income levels (<40k or ≥ 40k)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: K=12 • Rewiring probability: β=0.25

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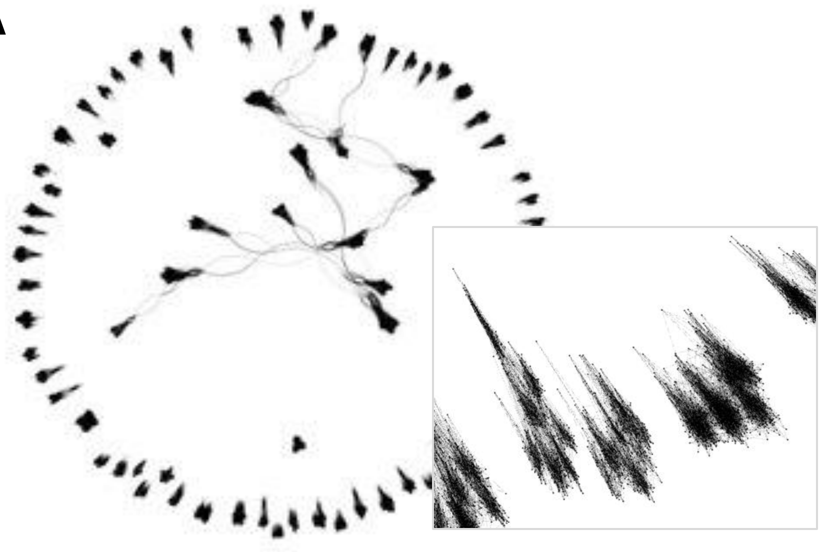
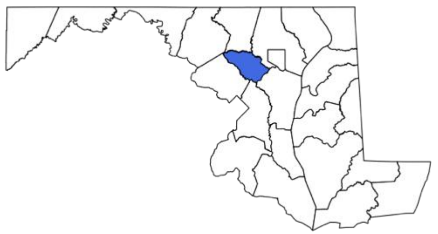
Network visualizations for Howard County, MD

A



B

p1	p2	beta
748	78	1
984	883	1
1702	115	1
1709	1521	1
1875	651	1



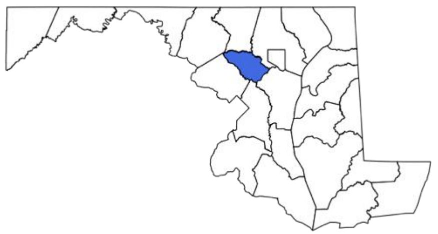
Visualization and table snippet of the school network. (A) Nodes are students and teachers clustered in grades within schools. (B) Same graph in Starsim tabular format where beta is the weight of the connection between person one and person two.

Code example

```
import geopops

pars = {‘path’: ‘YOUR_OUTPUT_DIR’,    # Where you want outputs
        ‘census_api_key’: ‘YOUR_KEY’, # Your Census API key
        ‘main_year': 2019,            # Main year
        ‘geos': [‘24027’],            # State/county FIPS of main geo
        ‘commute_states’: [‘24’],     # Commute to/from states
        ‘use_pums’: [‘24’]}           # PUMS states to download

geopops.WriteConfig(**pars) # Set parameters
geopops.DownloadData()      # Download raw data files
geopops.ProcessData()       # Preprocessing for next steps
geopops.RunJulia()          # CO, network generation, export to csv
geopops.ForStarsim()        # Format for Starsim
```



GeoPops with Starsim

Components passed into Starsim to model disease transmission

SIR model
- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

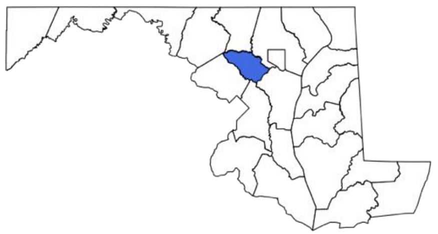
GeoPops agents

Feedback module
- Removes edges of school network when prevalence > 0.1

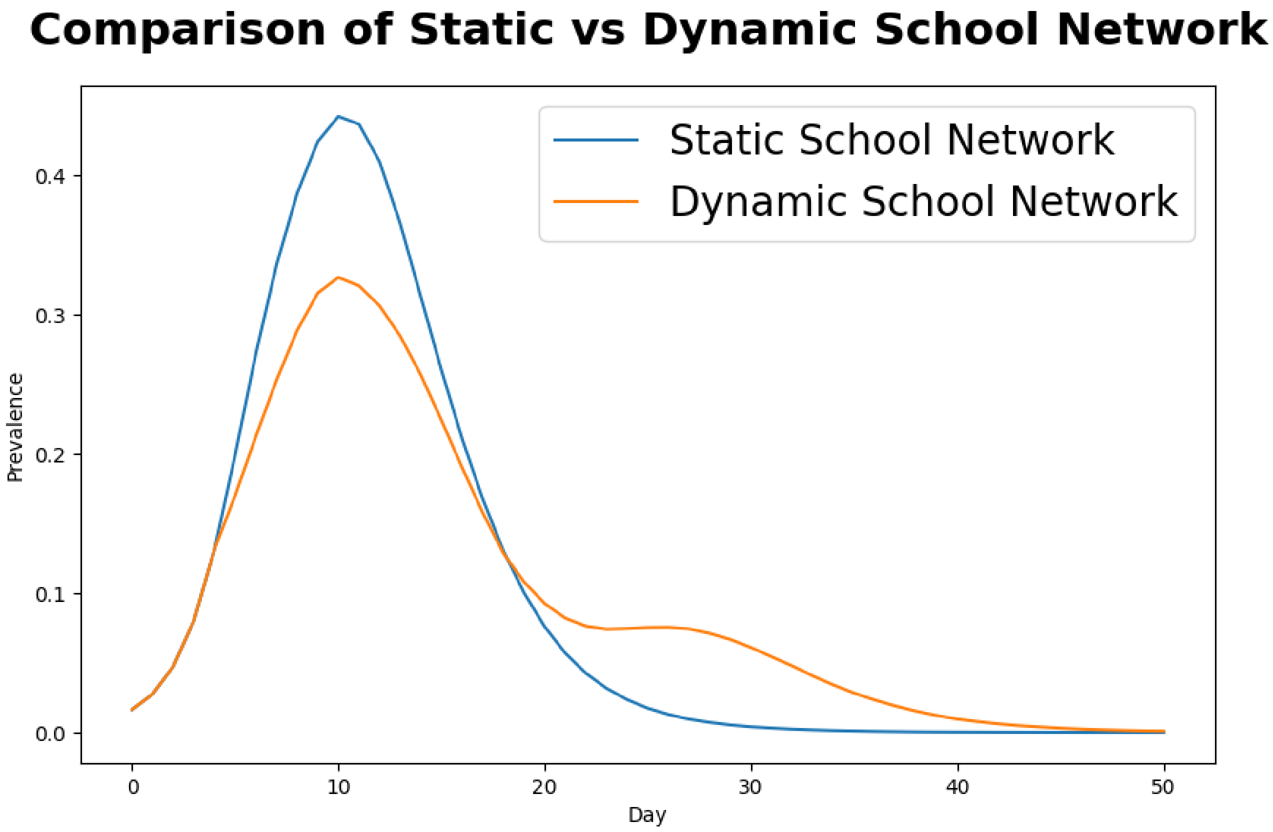
GeoPops networks

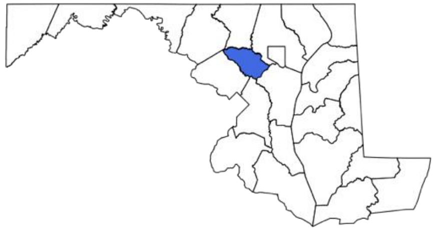
Subgroup Tracking
- Infections by age and CBG

Simple example with generic upper respiratory infection for Howard County, MD

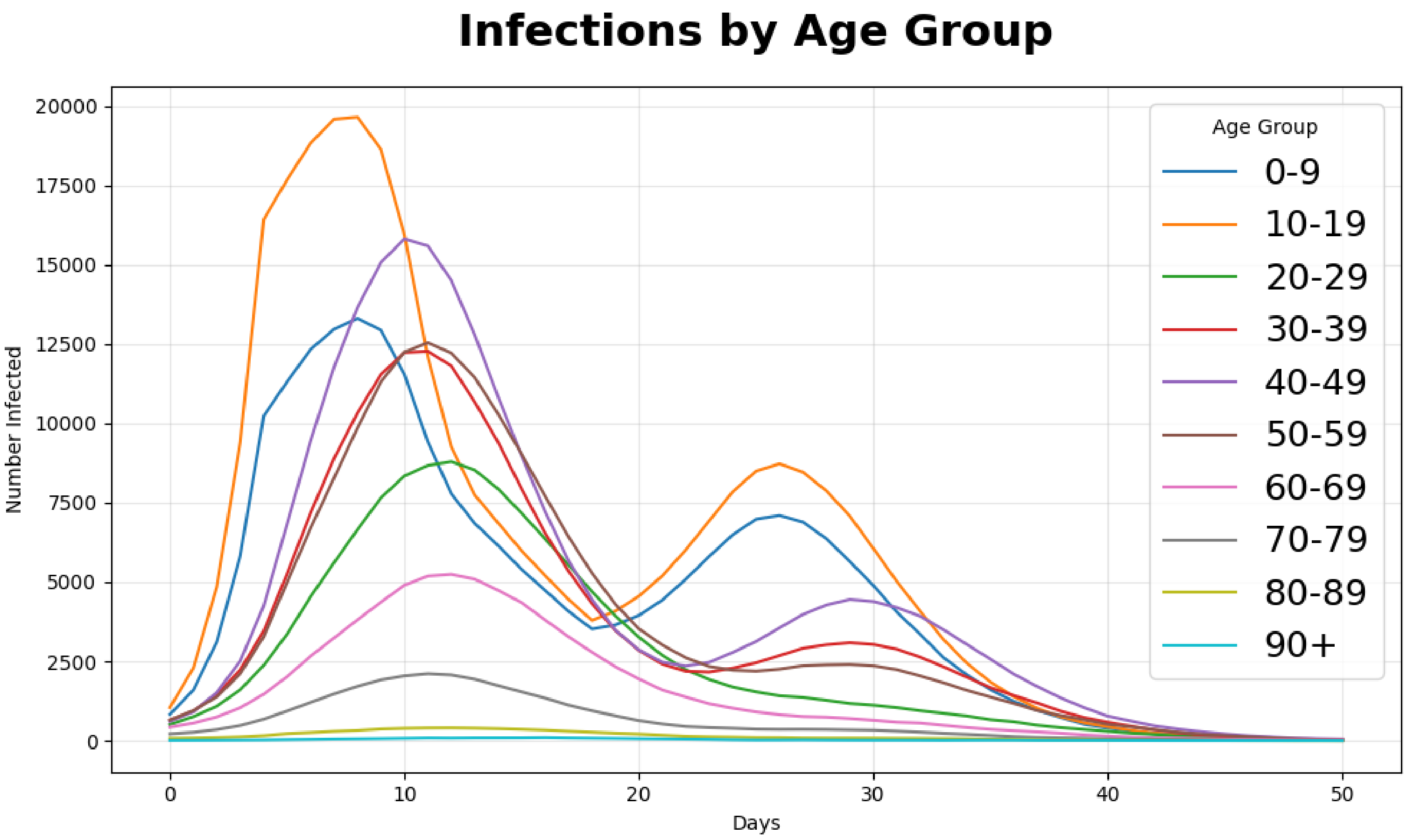


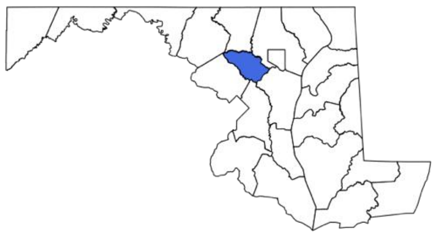
Feedback vs No Feedback



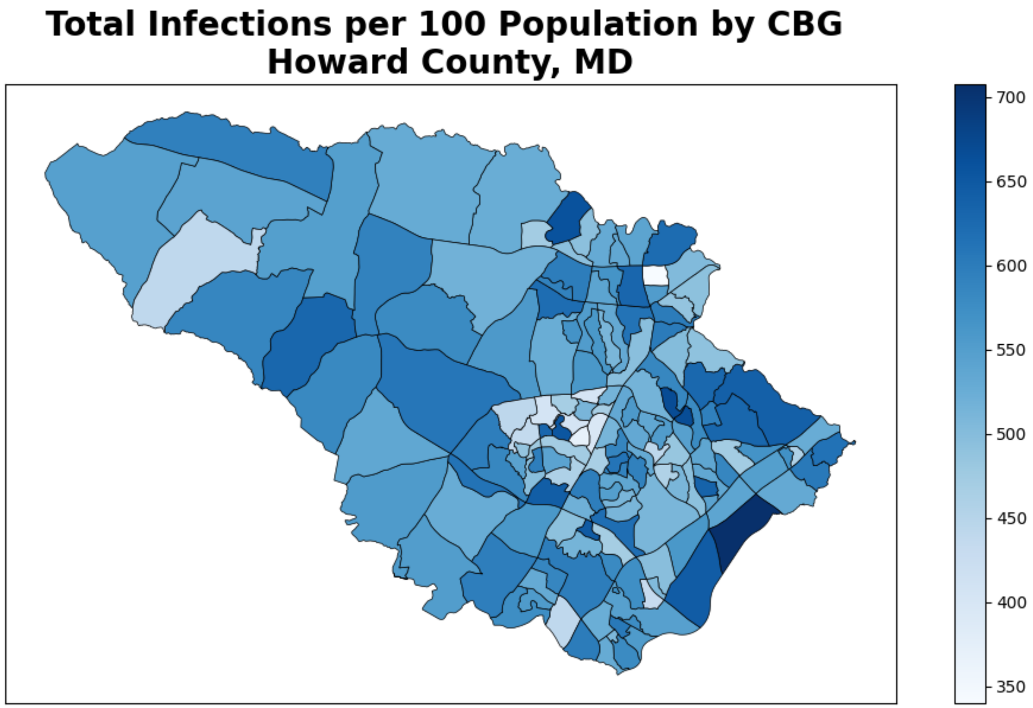


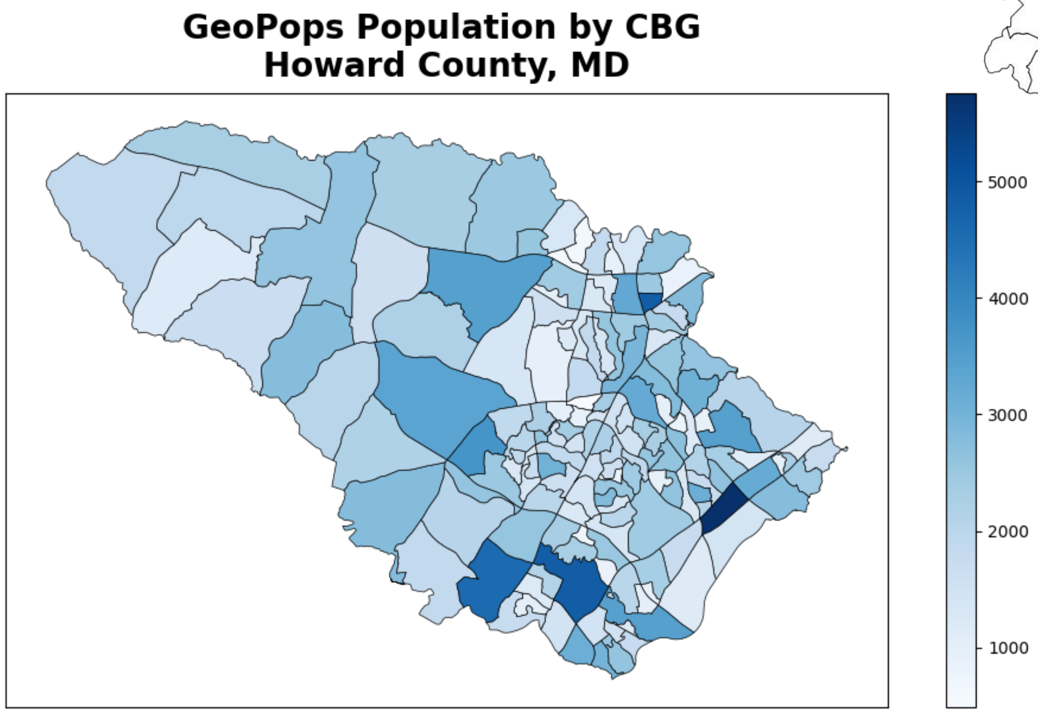
Outcome tracking by demographic subgroup

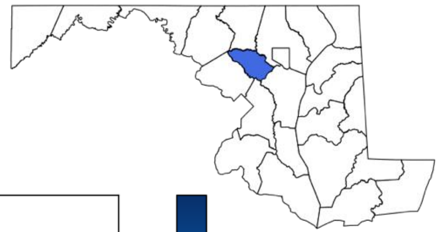




Outcome tracking by geographic subgroup







Why is this cool?!

Open-source, user-friendly framework for capturing demographic and geographic differences in human behavior and health outcomes.

- Don't have to build a model from scratch
- Can be used by state and county health departments for context-specific scenario modeling
- Seed infections to a specific geography or target interventions to specific demographic or geographic groups
- **Inform decisions about targeted interventions, which can lead to more efficient resource use and reductions in overall disease burden and disparities**
- There are several research groups already using Starsim to model respiratory diseases that can now use GeoPops as well

Actively maintained

Johns Hopkins Whiting School of Engineering

OneHealth Trust

ACCIDDA

IDM

University of Virginia

Johns Hopkins Applied Physics Laboratory

Conferences: Winter Simulation Conference

GeoPops: An open-source package for generating geographically realistic synthetic populations

- More detailed methodology and usage
- Compare GeoPops to two other open-source synthetic population generators (UrbanPop and Geo-Synthetic-Pop)
- Make three MD pops (one with each generator)
- Compare individuals and homes to real Census data
- Compare network stats of each home, school, and workplace network
- Run Starsim COVID-19 model on each population
- Compare simulation results to each other and observed data for first wave of pandemic
- Discuss usability of each generator

Conferences: Epidemics

Modeling the impact of dynamic decision making on infectious disease outcomes by demographic and geographic subgroups: An open-source agent-based modeling framework

- Run Starsim COVID-19 model with utility framework on GeoPops population of MD
- Individuals weigh health-wealth trade-offs of staying home from work as a function of their income and age
- Test different policy scenarios (e.g., paid sick leave, cash transfer) and compare to observed data from first wave of pandemic
- Demographic and geographic subgroup outcome tracking

Questions

- Pip install geopops and try going through the tutorial
- Log issues on GitHub! Please be patient ☺
- Poster, slides, and Notebook tutorial on GitHub

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github.com/ACCIDDA/GeoPops/tutorials/MIDAS

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